

Academic Background

semester number seven / starting my 4rd year mathematics major (2026-1) at Fundación Universitaria Konrad Lorenz, Bogotá Colombia, Interested in research.

Academic research

I am part of a research group led by Ph.D. José Alberto Vélez Marulanda, focused on Topological Data Analysis (TDA), where I have been working for one year. Our research explores applications of TDA methods on real datasets from Colombia. In my current project, we began studying flood risks in Colombia using a geometric big data framework. Beyond extensive data cleaning, we have implemented TDA techniques to analyze potential future flooded areas, regions at risk due to their topology, and possible relationships between zones. From the mathematical and topological perspective, we have worked with persistence diagrams, metrics, homology dimensions, Euler curves, and persistence landscapes. On the computational side, we employ parallel programming, vectorization, and comparative approaches to improve compilation time, using code partly in Python, with initial implementations in R, and future extensions planned in C++.

A portion of the code is available on my GitHub to promote open access, although I reserve the fundamental components for my research group.

I am currently working as a research assistant under Ph.D. José Alberto Vélez Marulanda in Topological Data Analysis. This work is directly connected to my undergraduate thesis, which focuses on applications of TDA combined with machine learning in financial domains.

Some parts of this project are also published on my GitHub to encourage free access to code, while I keep the core contributions reserved for academic purposes.

I developed a project for a psychology student that consisted of adapting a free artificial intelligence system to the point of challenging its morality, ethics, and human-like constraints, in order to debate sensitive topics such as abortion and gender. This project was mainly based on the architecture of artificial intelligence systems, modifying their APIs and then measuring the effectiveness of those changes.

If there is some interest in my research repository <https://github.com/nikomm13>

Areas of Expertise

- Advanced Mathematics: strong background in linear algebra, topology, statistics (regression methods, probabilistic models), and data analysis.
 - Topological Data Analysis (TDA): persistence diagrams, persistent homology, Euler characteristic curves, persistence landscapes, and metric-based approaches.
 - Programming and Computational Skills: proficient in Python and R, with experience in C++; parallel programming, vectorization, time complexity optimization, and algorithmic
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efficiency.

- Machine Learning and Data Science: application of supervised and unsupervised learning methods, integration of TDA with ML models, and analysis of large-scale datasets.
 - Research and Collaboration: strong investigative skills, teamwork experience in academic projects, and interdisciplinary applications of mathematics and computer science.
 - Open Science and Code Sharing: contributor to open-source projects via GitHub, balancing free access with preservation of core research contributions.
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Technical Skills

- Version Control: Git (branching, merging, collaborative workflows).
 - Programming Libraries: Python (NumPy, SciPy, scikit-learn, PyTorch), R (tidyverse, caret).
 - Computational Tools: Jupyter Notebooks, LaTeX, data visualization packages (Matplotlib, ggplot2).
 - Operating Systems: Linux (command line, shell scripting, environment configuration).
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Languages

- Spanish: native proficiency.
 - English: advanced reading, writing, and speaking skills.
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